

Workflow and architecture patterns for working together 2: Architectures

Jacob Archambault

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Outline

1 Current challenges and their canonical solutions

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- 2 Branching: theory and statistics
 - Theory
 - Statistics
 - Devops Statistics recapitulation
 - Approximating DevOps metrics in our current setup
 - Velocity
 - Safety
 - Maintenance to value-adding work percentage

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A naive probability theory for merge consistency: isolated commits

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A naive probability theory for merge consistency: modeling branches as sets of commits

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- $= \frac{1}{25}$

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A naive probability theory for merge consistency: general definition for distinct branches diverging from a common ancestor

- Let A be the set of unmerged commits across k distinct branches $A_1 \cdots A_k$, where $A = A_1 \sqcup A_2 \sqcup \cdots \sqcup A_k$, $n = |A|$, and $n_i = |A_i|$ for $1 \leq i \leq k$. Then

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$$P(\text{no conflicts}) = \frac{1}{2^n - (\sum_{i=1}^k 2^{n_i}) + k}$$

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The State of DevOps Report: throughput metrics

- lead time for changes

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- lead time for changes
- deployment frequency

The State of DevOps Report: stability metrics

- change failure rate

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- change failure rate
- mean time to restore

The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.'
- Nicole Forsgren, Jez Humble, and Gene Kim, Accelerate (2017), p. 14

The State of DevOps Report: results

When compared to low performers, elite performers realize

127x

faster lead
time

182x

more
deployments
per year

8x

lower change
failure rate

2293x

faster failed
deployment
recovery times

- 2024 State of DevOps Report

Approximating DevOps metrics in our current environment

- deployment frequency → GitHub Actions run count

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Velocity metrics

Branch	Runs	Earliest run	Contributors
NG_Claims_837D_COB	74	23 Mar	
PEDI-35120_837D_COB	85	15 Apr	5
Either COB branch	159	23 Mar	6
All other branches	123	23 Mar	12
	72	15 Apr	10
	266	21 Jan	

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- compare total commits across team branch vs. others
- compare time to merge across SpecFlow branches vs. others
- compare passing to failing pipeline runs across team branch vs. others, for both simulation pipeline and unit test pipeline
- compare non-merge commits to non-PR merge commits across team branch vs. others (this represents percentage of value-adding work to maintenance work) (`git log --oneline --merges | grep -iv`

“merge pull request” | wc -l)

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- Every change is tested by the test suite locally before committing

- tests are written against new code as that code is being written

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- daily PRs - stacked PRs

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- pipeline is definitive for deployment - daily stale branch job

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 - Testing
 - Pull Requests
 - Pipelines
- 4 Architecture